JACOBO MATEO BEDOYA OQUENDO

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SUMMARY

Working on production systems **since 2016**, in Python, with Go brought into the mix in 2022. Most of my work is optimization: making systems do more with less infrastructure, and keeping complexity from eating the codebase. Recent examples include Kafka and MQTT streaming for network observability at up to **1 million messages per second**, and request pipelines assembled from configuration instead of branching logic. Domains: carrier networking and telemetry, IoT and sensor pipelines, distributed storage, usage-based billing, real-estate investment analysis and industrial monitoring. More of the problems I have worked through are documented at jacobobedoya.com, in English and Spanish.

CORE TECHNOLOGIES

Python (since 2016) — FastAPI, Django, Django REST Framework, Flask, Celery, multiprocessing, metaprogramming
Go (since 2022) — services, CLI tooling, high-throughput pipelines · **C** — basic
Messaging & streaming — Kafka, MQTT (pub/sub), gRPC, WebSockets, event-driven pipelines
Datastores — PostgreSQL, TimescaleDB, ScyllaDB, Redis, MySQL, SQLite, MongoDB
Cloud — AWS (S3, Lambda, Elastic Beanstalk, EC2), GCP (Cloud Run, BigQuery, Cloud Storage), Azure (Blob Storage)
Infrastructure — Docker, Docker Compose, Docker Swarm, Kubernetes (exposure)
Architecture — event-driven, publisher–subscriber, client–server, peer-to-peer, service-oriented
Packaging & dependencies — uv for virtual environments and dependency management, lock-file based reproducible installs, Docker multi-stage builds
Code quality & CI — ruff, mypy, pre-commit hooks, CI-gated checks, JSON-driven automated test generation, DependaMan (own tool) for import-cycle and coupling analysis
Spoken languages — Spanish (native), English (C2, EF SET 82/100)

WORK EXPERIENCE

SHUSH — January 2025 – Present

Senior Backend / Platform Engineer (Contractor) — Remote

Contract role: part-time, a few hours per month, from January 2025 to December 2025; full-time since January 2026. Own the design and implementation of the platform's products end to end, translating requirements from clients and owners into delivered services, and built and maintained the platform backend in **Python/FastAPI** and its **gRPC micro-services**.

- Designed and implemented the platform's **configuration-driven request pipelines**: each endpoint's chain of pre-checks (authentication, permissions, rate limiting) and wrappers (logging, error handling) is declared in a TOML file and assembled at startup through runtime attribute mutation. Per-endpoint and per-deployment behaviour changes by configuration alone — no conditionals in endpoint code, no dead branches per request, no redeploy to change which checks run where.
- Built and maintained the **dynamic usage-based billing system** that meters and charges every client on the platform.
- Designed and delivered **network observability and session-lookup tooling** in Go for the CGNAT infrastructure of multiple carriers, covering up to **10 million subscribers**, built as two tools at two scale points so each carrier runs — and pays for — only the infrastructure its traffic actually needs:
  - a single-process, pipeline-based tool processing up to **150,000 messages per second**;
  - a **Kafka**-backed, event-driven micro-services tool processing up to **1,000,000 messages per second**.
- Packaged per-carrier releases with **multi-stage Docker builds**, so each image ships only the code generated for that carrier, without the build toolchain or the sources behind it.
- Enforced code quality with **ruff and mypy via pre-commit hooks and CI-gated checks** across the Python services.

Stack: Python, FastAPI, gRPC, Kafka, ScyllaDB, Go, SQLite, Docker, event-driven micro-services

EarthFuneral — March 2025 – December 2025

Senior Backend Engineer — Remote

Real-time monitoring platform for a fleet of around **1,000 machines**. Developed and maintained its **Python/Django** backend, shipping features and improving overall stability.

- Built a **real-time sensor data pipeline** with MQTT (pub/sub), WebSocket broadcasting, Redis caching and asynchronous task execution via Celery, sustaining **150–160 sensor messages per second**.
- Modelled each composting machine as a **digital twin** that operators drive from the web platform, with a **state watcher** underneath reconciling the physical equipment against the virtual one and correcting it when the two diverged.
- Implemented a **multi-database architecture** — PostgreSQL for transactional data, TimescaleDB for time series — behind a custom database router.
- Delivered an **alert-response system**: detects equipment failure over MQTT, triggers automated actions through Celery and pushes immediate notifications to monitoring dashboards over WebSocket.
- Improved performance through query optimization, indexing and Redis caching, **cutting database usage from 60% to 45%**.

Stack: Python, Django, Celery, MQTT, WebSockets, Redis, PostgreSQL, TimescaleDB

Endava — August 2022 – December 2024

Senior Developer (Engineer) — Bogotá, Colombia

Math Engine — Real Estate Loan Analysis (June 2024 – December 2024)

- Built a loan-sizing and investment-analysis engine; implemented **JSON-driven automated test generation** covering every scenario, and multi-objective binary search over the optimization space to solve for maximum loan amount and minimum net operating income.

- Published the open-source Go version of the package.

Point Cloud — Desktop Annotation Tool (August 2022 – May 2024)

- Built and maintained the **PyQt** desktop application used to annotate point clouds through a graphical interface, with **VTK** for 3D rendering.
- Led the Python side of the **migration from Python 3.2 to Python 3.10**, with a teammate leading the **PyQt6** and **VTK** moves, including reworking the event system onto messages and signals. A major refactor across a compiled, GUI-heavy stack, reconciling behaviour changes and dependency constraints.
- Led the design and implementation of an **event-driven architecture** for compute-heavy features, distributing load across cores with multiprocessing and reducing object-creation overhead: **double the speed and 40% less RAM**.
- Automated blueprint and technical-drawing generation via a local server reacting to annotation changes in the point cloud.
- Built a thread-safe **facade over the BOX Cloud SDK** for file updates and uploads.

Stack: Python 3.2 → 3.10, PyQt6, VTK, multiprocessing, Go, event-driven architecture

Sophos Solutions — April 2022 – July 2022

Software Developer — Medellín, Colombia

Continued the development and scaling of a mine monitoring platform for geological risk mitigation.

- Built and maintained the platform's **Django REST Framework** backend, consuming mine sensor events delivered over **Kafka**.
- Developed the **mine-local Flask services** that streamed sensor data from each site up to the platform.
- Documented the inherited codebase and resolved the **Kubernetes** infrastructure problems that were blocking delivery.
- Acted as the technical bridge between the Spanish-speaking team that built the platform and the English-speaking team taking it over, which unblocked delivery to the client.

Stack: Python, Django REST Framework, Flask, Kafka, Kubernetes

ActiveOne — May 2021 – April 2022

Development Analyst — Medellín, Colombia

- Backend development in **Python/Django** across the company's client portfolio, including a web platform for uploading and classifying scanned proceedings through AWS AI services.
- Built the **streaming transfer tool** behind a cloud storage migration, moving files from **AWS S3** to **Azure Blob Storage** as a stream from one service to the other, without staging to disk.
- Implemented an application-wide **logging framework** from a base metaclass plus function decorators, instrumenting critical paths with a one-line change.

Stack: Python, Django, AWS (S3, AI services), Azure Blob Storage

LUJACOL S.A.S. — March 2020 – June 2022

Owner & Webapps Developer — Medellín, Colombia

Personal company, founded and run alongside my employed roles, and later sold. I built the software the operation ran on, including the client portal used to place orders, follow shipping status and download the documents needed for merchandise nationalization.

- Built the **inventory management system** and a **delivery calendar that synced dynamically with the Hapag-Lloyd carrier**, tracking the vessel carrying each order.
- Automated client communications — email reminders, ISO documentation expiry alerts, price calculation — which kept headcount minimal and container preparation under a week.

Stack: Python, Django, SQLite

Institución Universitaria ITM — October 2016 – January 2021

Data Analyst — Medellín, Colombia

- Built the web platform for academic follow-up of every student on a Medellín mayor's office scholarship, generating early alerts on low performance and dropout risk to trigger student-welfare intervention.
- Designed a metric quantifying the university's academic impact per student across performance tiers, used to evaluate student-retention programs.

Stack: Python, MySQL — first professional Python work (2016)

PUBLIC PROJECTS

- **JaDFS** (November 2025 – present) — distributed file storage system in Go, evolving from a single file server to a fully distributed, fault-tolerant storage system. codeberg.org/jacobitosuperstar/JaDFS
- **DependaMan** (March 2026 – May 2026) — Python dependency and static analysis tool: maps a project's internal module structure and surfaces import cycles, coupling issues and hotspots as an interactive HTML graph. codeberg.org/jacobitosuperstar/DependaMan

EDUCATION & CERTIFICATIONS

M.Sc. Automation and Industrial Control — Institución Universitaria ITM, January 2017 – January 2022. Coursework completed; unfinished thesis on spectrum optimization for data streaming.

Mechanical Engineering — Universidad Pontificia Bolivariana, January 2008 – April 2016. Control systems and state observability — the background behind the industrial monitoring, sensor and equipment-state work above.

Certifications — gRPC Master Class · Generative AI for Developers · EF SET English Certificate — 82/100 (C2 Proficient)